

Points of intersection



1

a 2

b 1

c 0

d 1

e 0

f 2

g 1

h 1

i 0

j 0

2 $(3, 4), (-2, 9)$

3 Minimum value of $k = 1$

4

a $4x^2 - 5x + 14 = 0$

b -199

c No, because the discriminant is negative which means the quadratic has no solution.

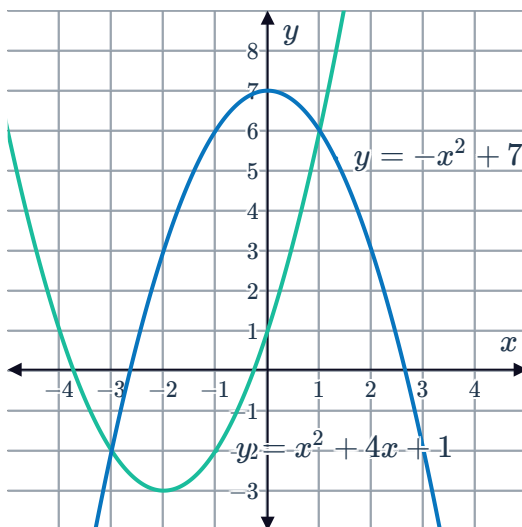
5 5

6 $(-5, -67)$

7 $k = 7$

8

a
i

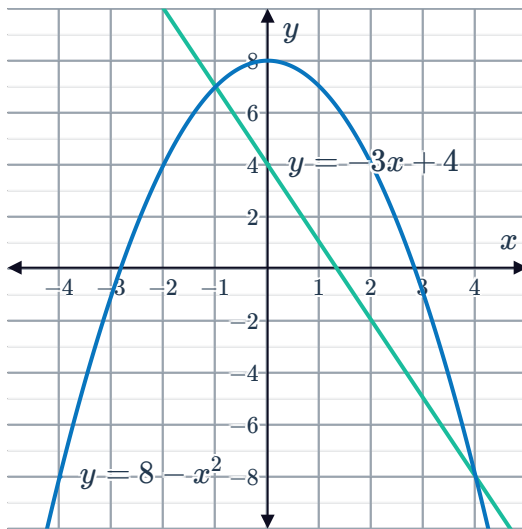


ii $(1, 6), (-3, -2)$

b
i



ii $(-1, 7), (4, -8)$



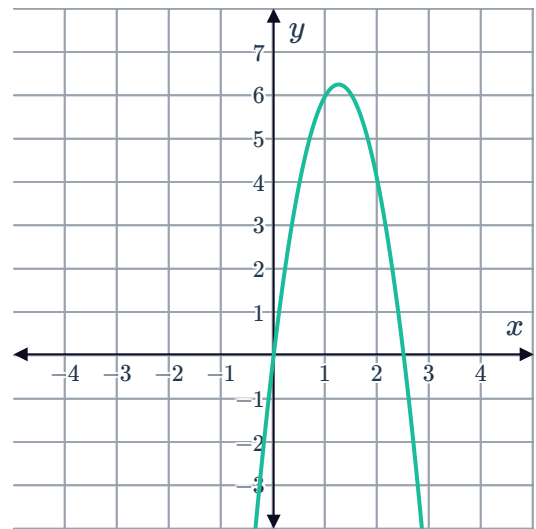
9

a $x = 0, \frac{5}{2}$

c $\left(\frac{5}{4}, \frac{25}{4}\right)$

b $x = \frac{5}{4}$

d



e $c = 6.25$

f $c < \frac{25}{4}$

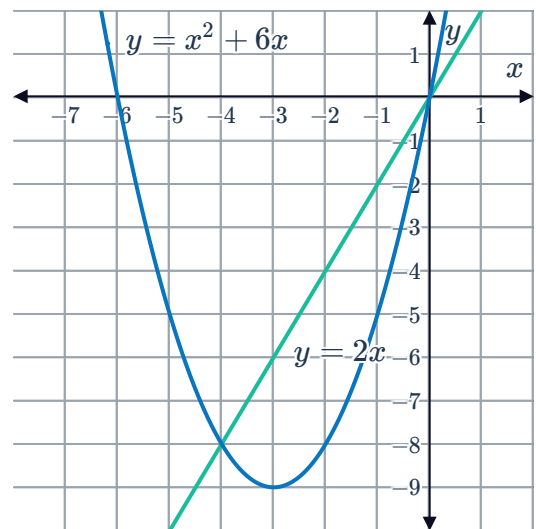
10

a

x	-4	-3	-2
$y = x^2 + 6x$	-8	-9	-8
$y = 2x$	-8	-6	-4



b

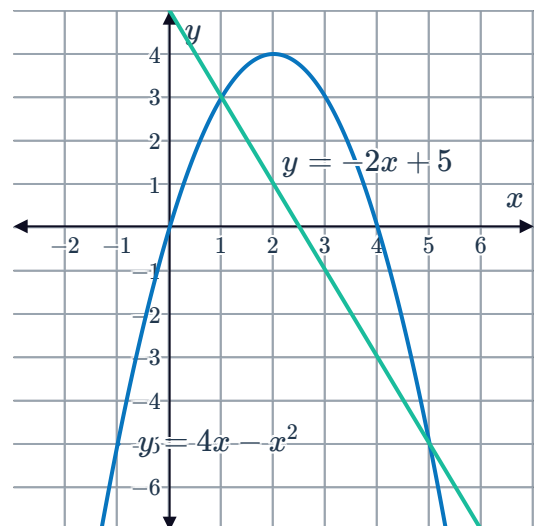
c $(0,0), (-4,-8)$

11

a

x	1	2	3
$y = 4x - x^2$	3	4	3
$y = -2x + 5$	3	1	-1

b

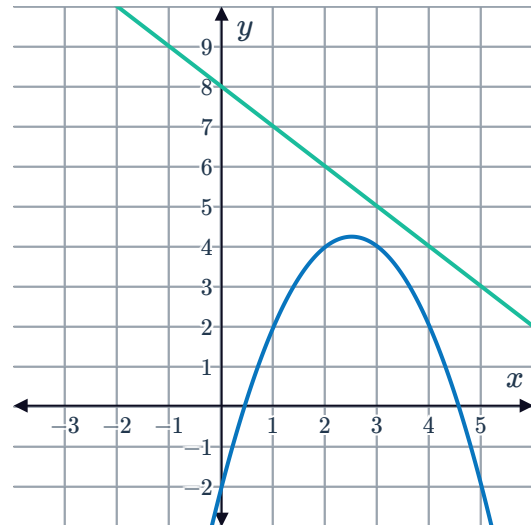
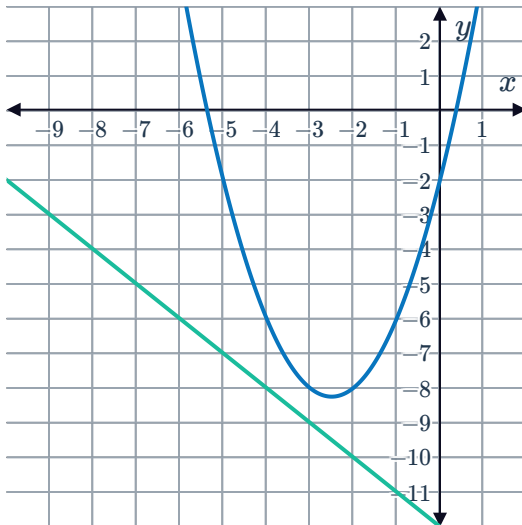
c $(1,3), (5,-5)$

12

a No

b $(-2,7)$

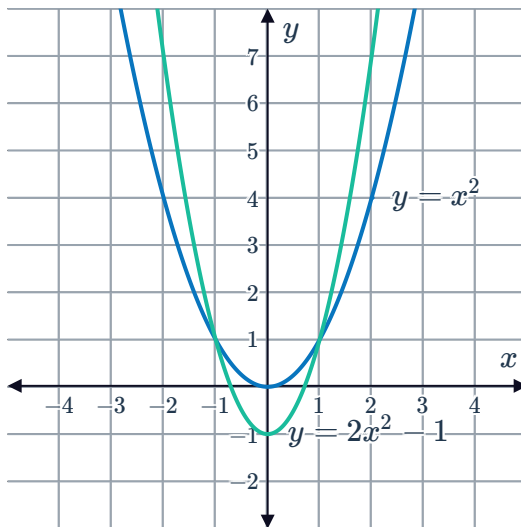
13 Example answers:



14

a

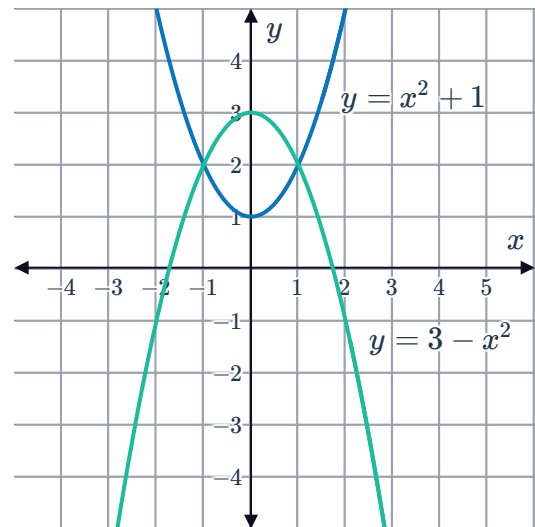
i



ii $(-1, 1), (1, 1)$

b

i



ii $(-1, 2), (1, 2)$



Quadratic systems of equations

15

- | | | | | |
|---|---|--------------------|----|---|
| a | i | $x = \pm 2$ | ii | $(-2, 0), (2, 0)$ |
| b | i | $x = \pm 2$ | ii | $(-2, 4), (2, 4)$ |
| c | i | $x = 0$ | ii | $(0, 0)$ |
| d | i | $x = 3, -5$ | ii | $(3, 25), (-5, 9)$ |
| e | i | $x = 7$ | ii | $(7, 8)$ |
| f | i | $x = 5, 4$ | ii | $(5, 54), (4, 38)$ |
| g | i | $x = \frac{1}{4}$ | ii | $\left(\frac{1}{4}, -\frac{109}{16}\right)$ |
| h | i | $x = \frac{21}{2}$ | ii | $\left(\frac{21}{2}, \frac{5}{4}\right)$ |
| i | i | $x = 2$ | ii | $(2, 6)$ |
| j | i | $x = \pm\sqrt{10}$ | ii | $(-\sqrt{10}, -8), (\sqrt{10}, -8)$ |

16

- | | | | | | | | |
|---|--------------------|---|---------------------|---|--------------------|---|---|
| a | $(0, 0), (-7, 49)$ | b | $(3, 18), (-7, 28)$ | c | $(3, -7)$ | d | $(0, 0), (1, 16)$ |
| e | $(7, -7), (1, -7)$ | f | $(-4, 42)$ | g | $(2, 2), (-3, -3)$ | h | $(3, 0), \left(\frac{7}{2}, \frac{1}{4}\right)$ |
| i | $(2, 1), (-2, 1)$ | | | | | | |

17

- | | | | |
|---|--|---|--|
| a | $x = \frac{1 + \sqrt{17}}{2}, \frac{1 - \sqrt{17}}{2}$ | b | $\left(\frac{1 + \sqrt{17}}{2}, -4\right), \left(\frac{1 - \sqrt{17}}{2}, -4\right)$ |
|---|--|---|--|

18 $(0, 16), (1, 15), (-1, 15), (-2, 16)$



19

a $(-2, 3), (2, -9)$

c $(2, -4), (0, 0)$

b $(-4, -12), (-3, -12)$

d i $x = 4, -3$ ii $(4, 0), (-3, 21)$

e i $x = -4, -5$ ii $(-4, 2), (-5, 5)$

20 $y = -x - \frac{21}{4}$

21

a $x = -4 \pm 2\sqrt{3}$

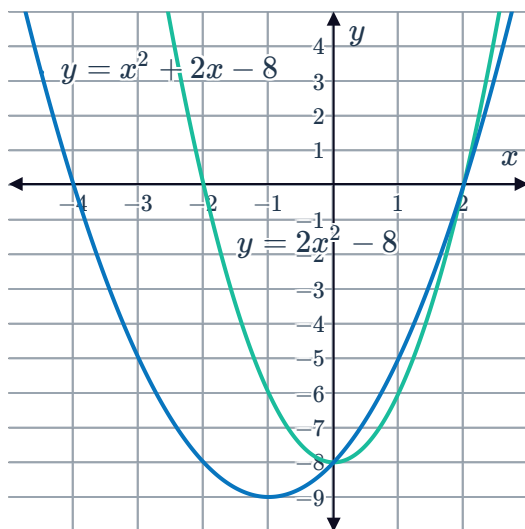
b i Yes ii Yes iii Yes

22

a $c = -8$

b $a = -2$

c



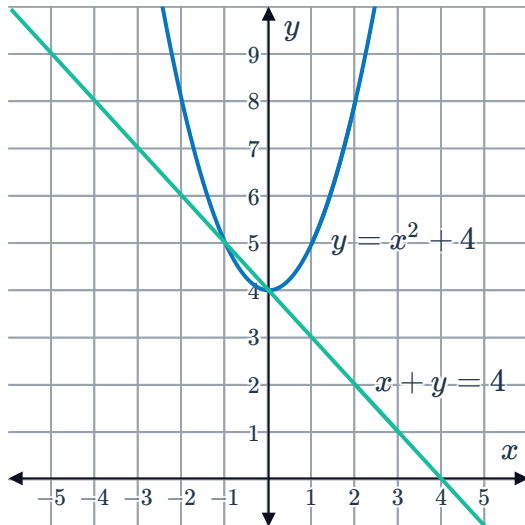


Quadratic equations from systems

23 $3x^2 - 2x + 9 = 0$

24

a



b 2

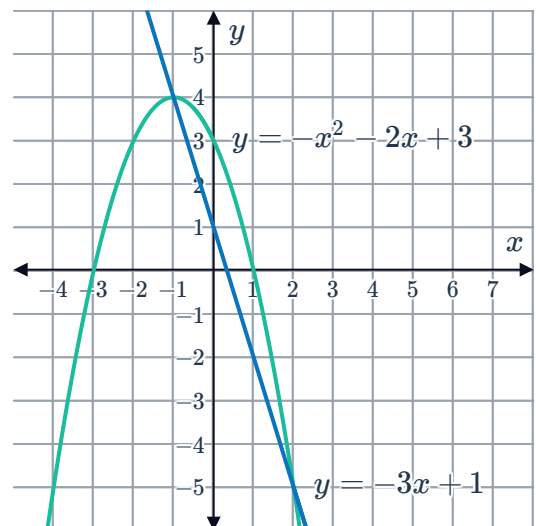
c $(0, 4), (-1, 5)$

25 $y = 2x + 6$

26

a $-3x + 1$

b



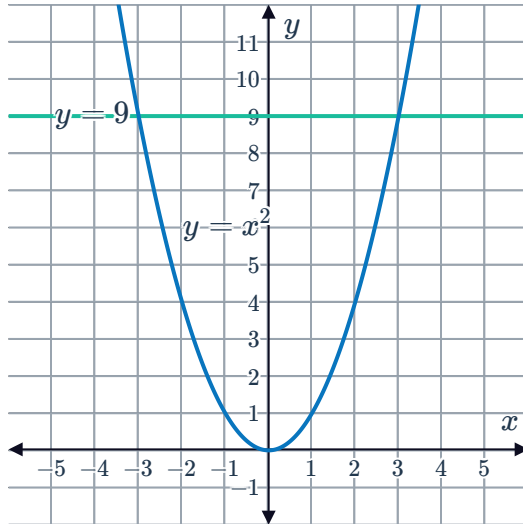
c $x = -1, 2$

27



28

a



b $(-3, 9), (3, 9)$

c $x = \pm 3$

Applications

29 $y = 1, \frac{8}{3}$

30

a $x = 0, x = 6$

b $y = 144$

c Example answer:

The equilibrium quantity occurs when 6 thousand units are being produced, which has a corresponding price of 144 dollars per unit.

